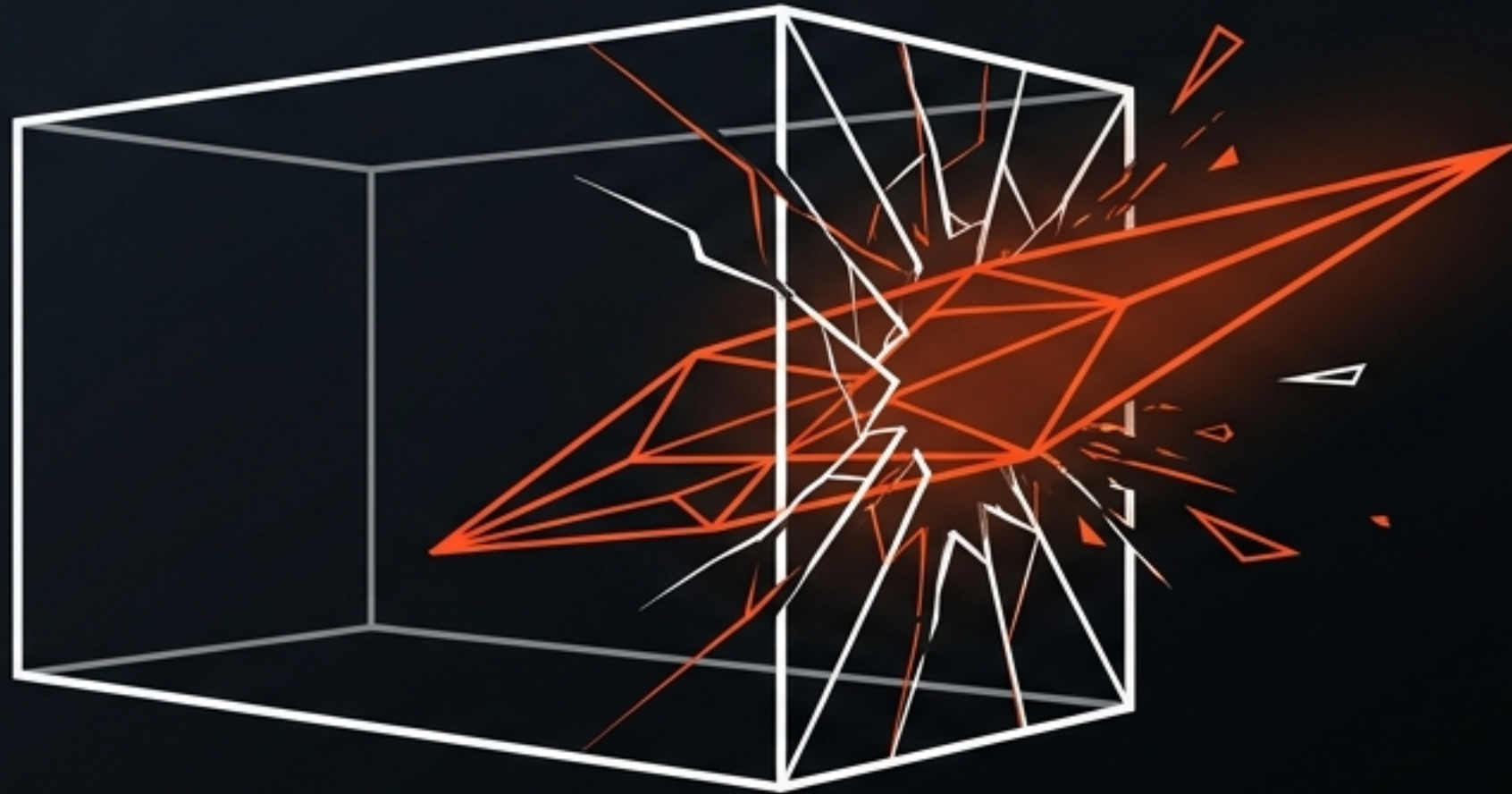


Beyond Prediction Error

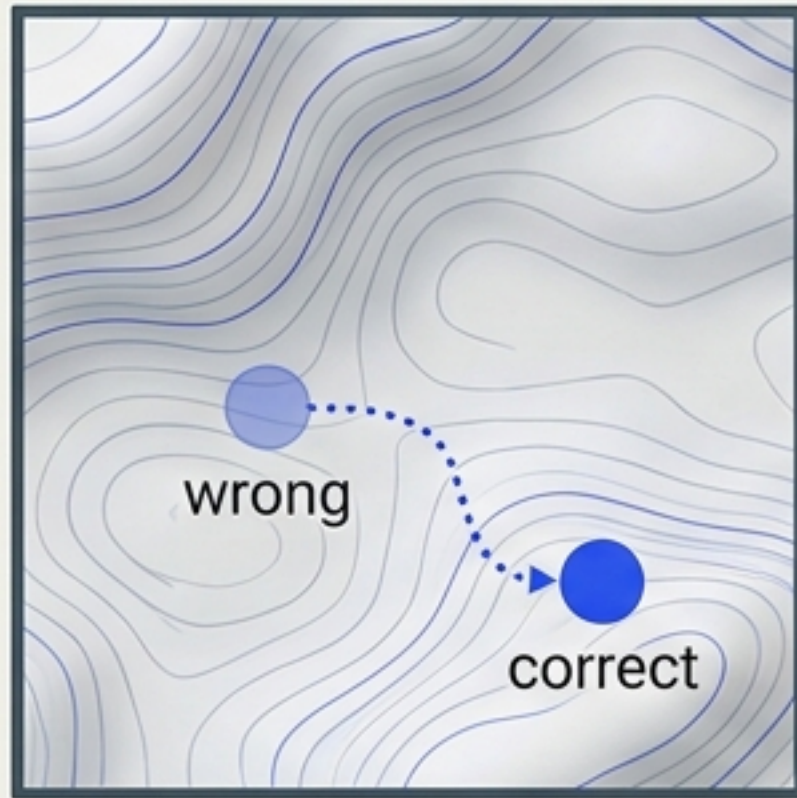
Warrant, Repair, and Refusal in Predictive Architectures

Why inference within a space is not equivalent to revision of the space.



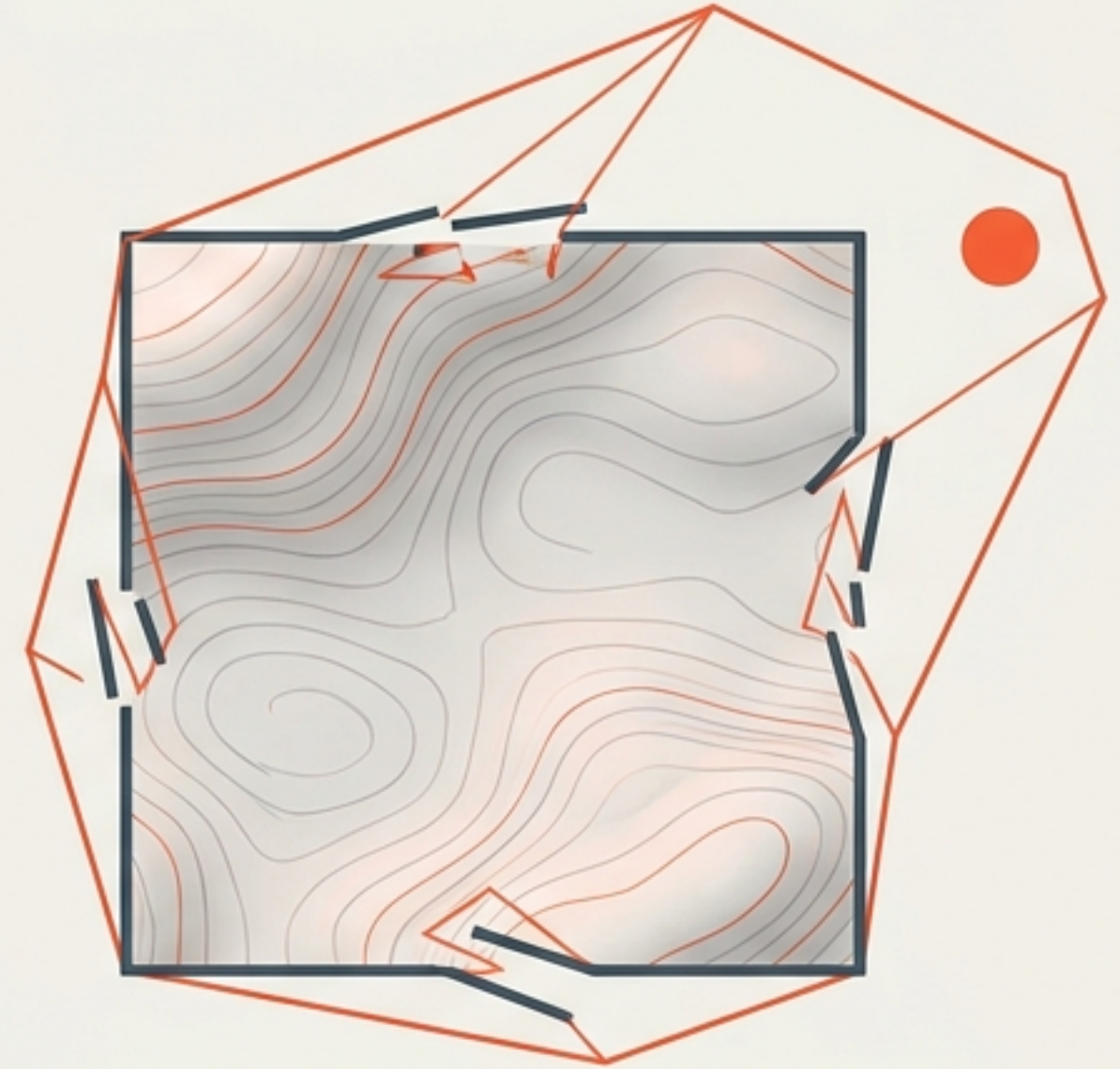
Based on the 2026 paper by Flyxion.

A system can be wrong in two fundamentally different ways.



1. Error Within a Space.

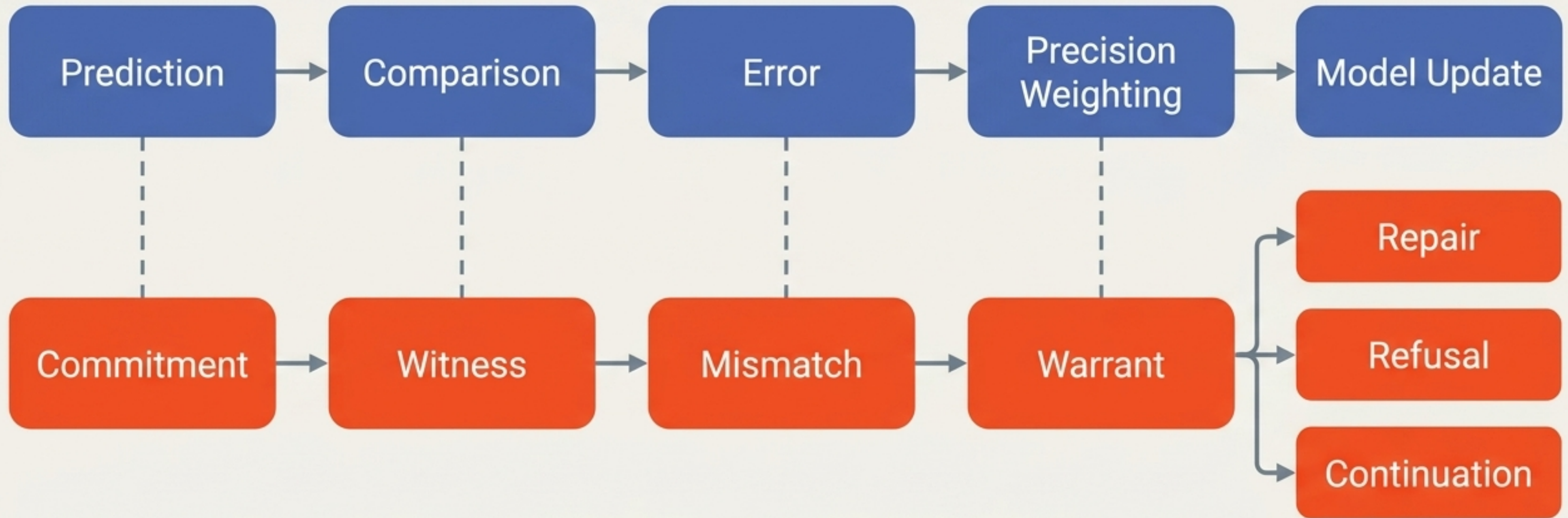
The system occupies the wrong position in its current space of possibilities. This is an error for inference to correct.



2. Error About a Space.

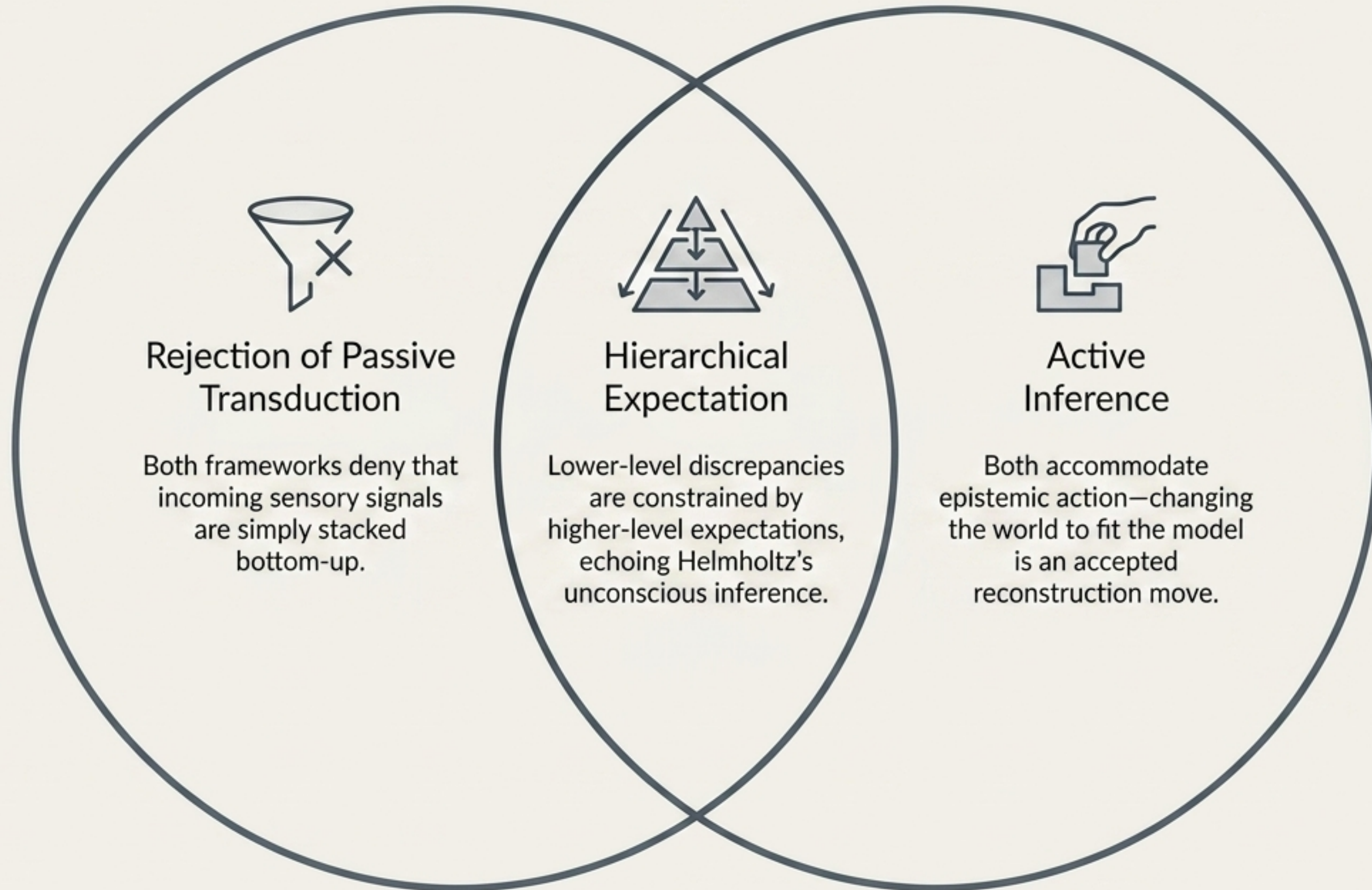
The system is operating with the wrong space of possibilities entirely. This is an error about what inference is permitted to be.

The Illusion of Equivalence



The two architectures look like notational variants. The analogy holds perfectly—until the final outcome.

Where the Architectures Genuinely Agree

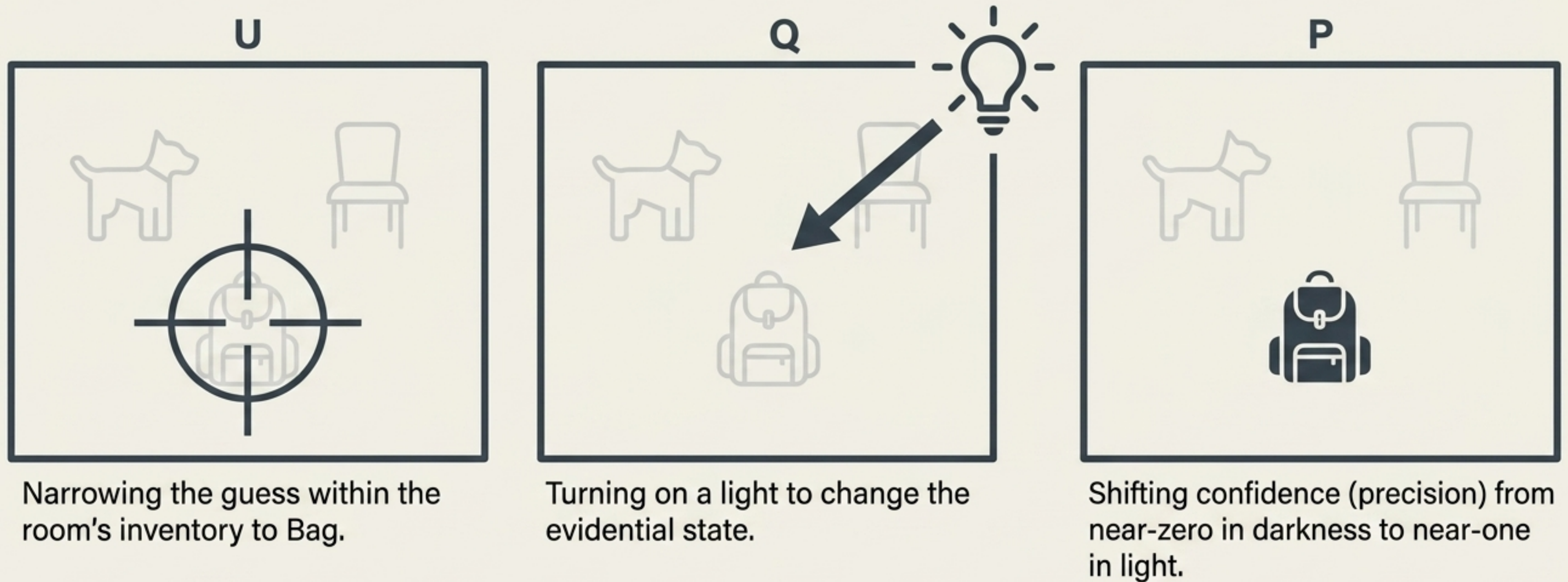


The Typological Fracture: Four Distinct Operators

U Update	Q Query	P Precision	A Admissibility
Type: Type: $U : X \times \Omega \rightarrow \Omega$ Effect: Revises a commitment within the current admissibility space.	Type: Type: $Q : X \times \Omega \rightarrow E$ Effect: Changes the evidential state available to inference (epistemic action).	Type: Type: $P : W_{\Omega} \rightarrow W_{\Omega}$ Effect: Alters the inferential influence assigned to discrepancies (reweighting).	Type: $A : D \rightarrow D, \Omega \mapsto \Omega'$ Effect: Changes the admissibility space itself.

Navigating the Known Space

An unfamiliar shape on the floor of a dark house. The admissible space (Ω) contains normal household objects.



Throughout U, Q, and P, the space of possibilities (Ω) remains completely unchanged.

Confronting the Unprecedented (A)

The light reveals something entirely outside pre-existing categories, or an alibi definitively rules out an existing category.

A+ Admission



A+ (Admission): $\Omega \cup S$

The mismatch cannot be resolved by guessing better or trusting evidence more. The space itself must be extended.

A- Refusal



A- (Refusal): $\Omega \setminus R$

An established hypothesis is not just down-weighted; it is constitutively removed from the admissible set.

The Mathematical Proof: Reweighting $\neq$ Refusal

Reweighting (P)

$$P(w)(r) = 0$$



Assigning absolute zero weight to a hypothesis leaves it inside the domain. The codomain is always W_Ω .

Refusal (A-)

$$A-R(\Omega) = \Omega \setminus R$$

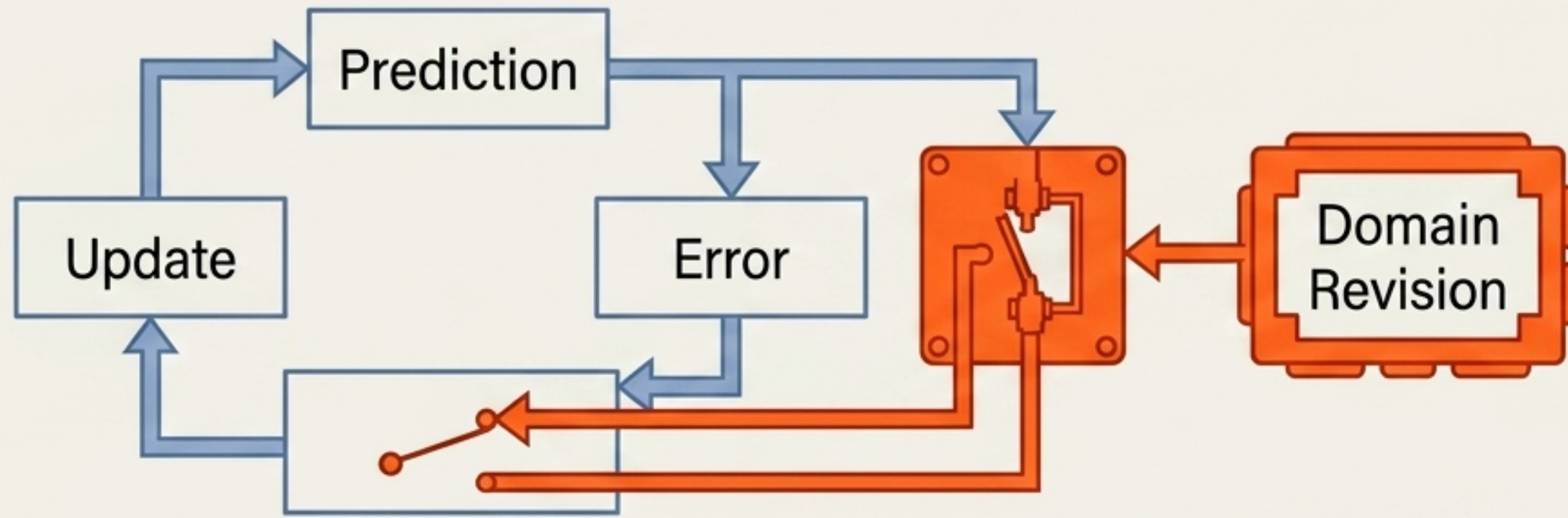


The element is removed; weights are subsequently defined on a fundamentally different space.

Theorem: No precision reweighting operator (P), even pushed to absolute zero, can instantiate constitutive refusal (A-). Zero is a value assigned to an element that remains in the domain.

Structural Additions for Predictive Architectures

Predictive architectures aren't internally inconsistent, but they require additional **structure** to represent **space revision**.



Solution 1: The Commitment Variable (c)

A Boolean mechanism $c : H \rightarrow \{0,1\}$
where changing a state $1 \mapsto 0$ automatically
updates the admissible space $\Omega_c \mapsto \Omega'_c$.

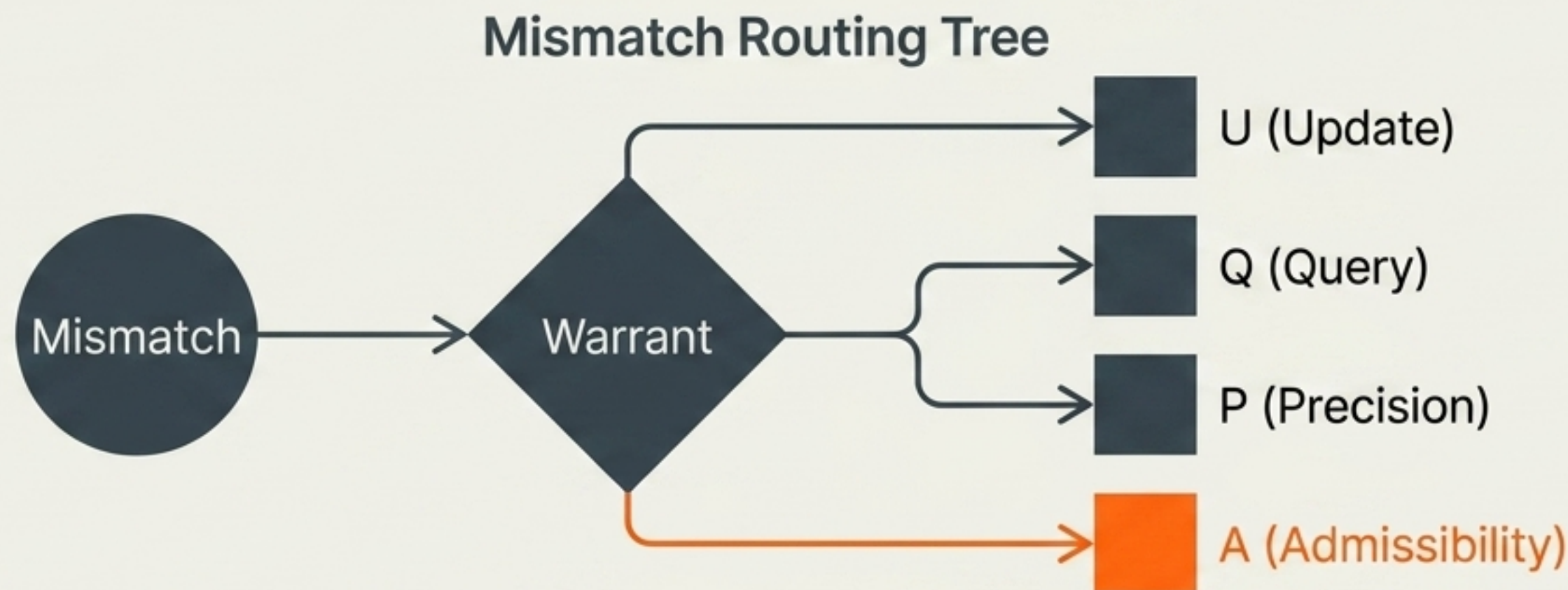
Solution 2: Dynamic Constraint Sets (C)

Altering the underlying constraint rules $C \mapsto C'$
forcing a re-evaluation of which hypotheses
satisfy the new constraints.

Core Requirement: The architecture requires some state or operator whose variation changes membership in the space over which inference is performed.

The Warrant Problem

Error correction $\neq$ evidence acquisition $\neq$ domain revision.



Mismatch alone does not determine its remedy. A witnessed event can cause a repair without being a repair. A system requires a theory of Warrant to dictate whether a discrepancy means we should change our guess, seek more evidence, alter our trust in the data, or fundamentally restructure the domain of possibility.

Clarification: Structure Learning is Not Domain Revision

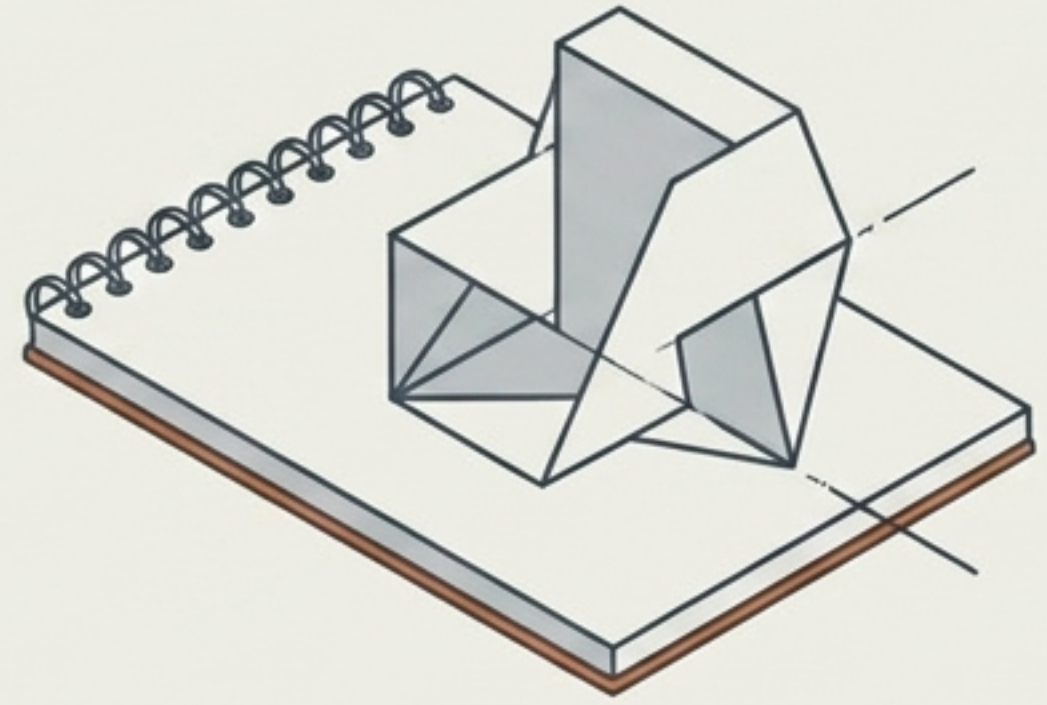
Structure Learning



The Objection: Nonparametric models grow the hypothesis space as data is observed.

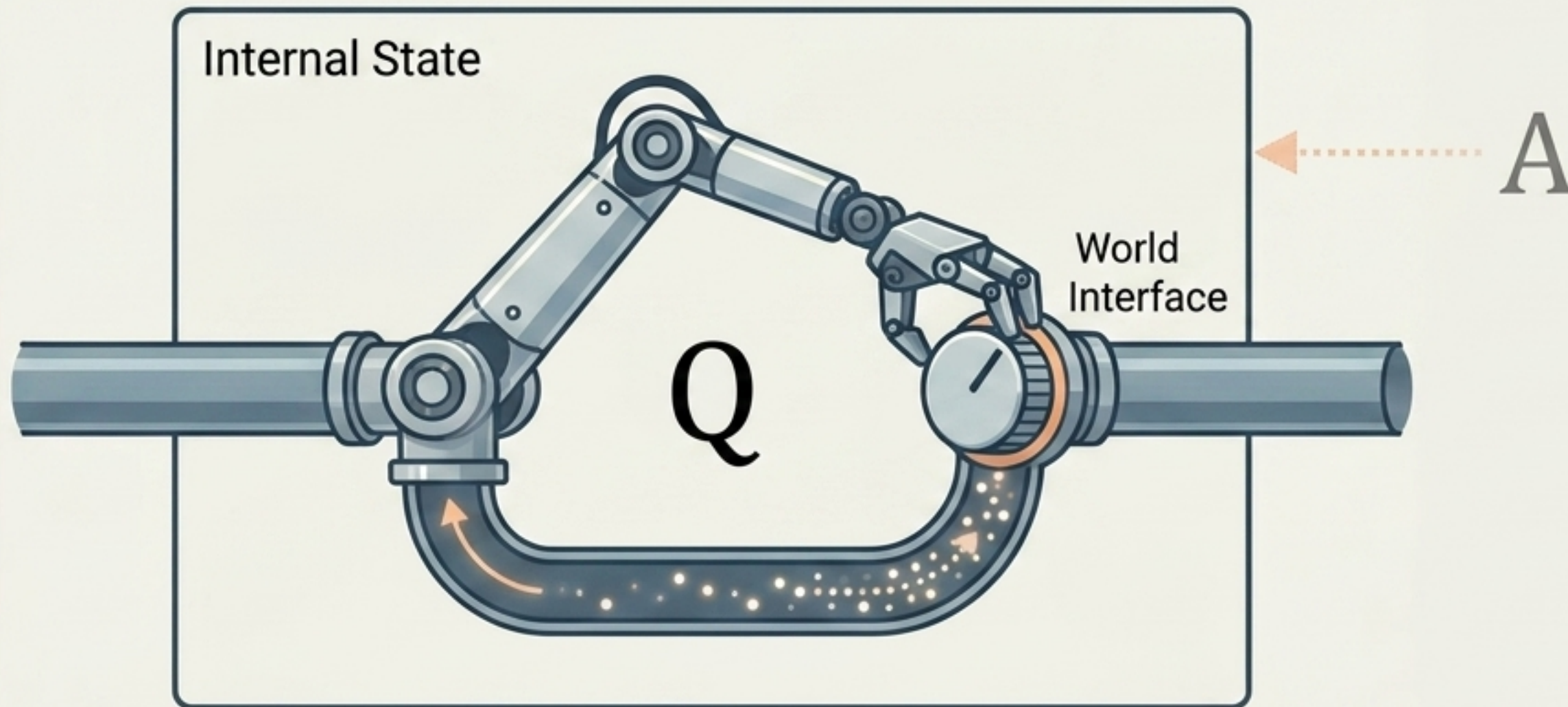
The Reality: This is domain growth drawn from a prior fixed in advance. It is a highly sophisticated instance of U or P operating one level up.

Constitutive Revision



The Distinction: Real admissibility revision (A) is triggered by a witnessed mismatch that the model's designer did not anticipate, relying on an external warrant condition rather than a predetermined mathematical prior.

Clarification: Active Inference Maps to Operator Q



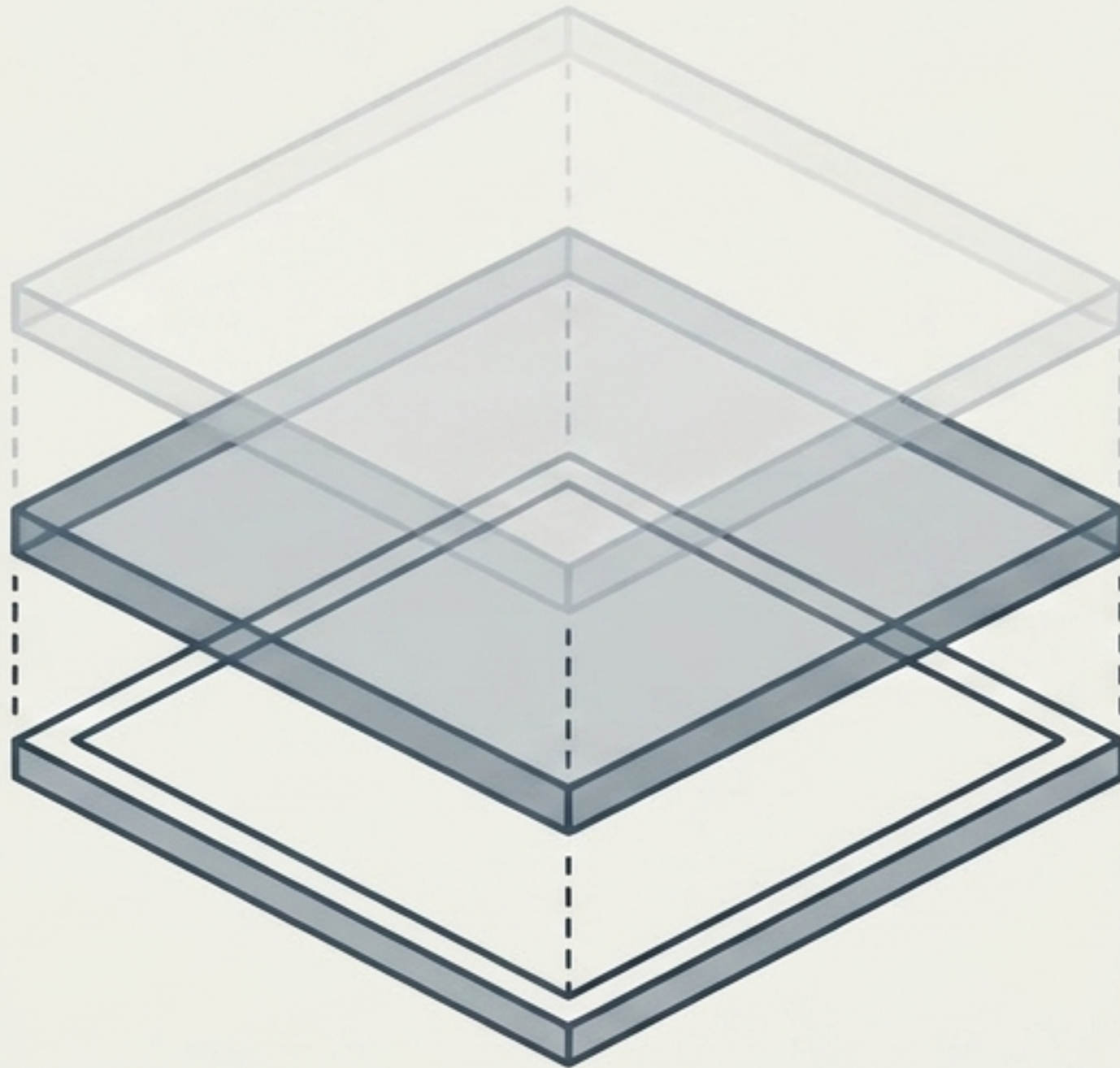
The Objection: Active inference models can change their encounters and, in principle, their generative models.

The Reality: Epistemic action changes the evidential state available to inference—this is a textbook implementation of Q.

The Synthesis: If an active inference architecture also revises its generative model's structure (not just parameters) in response to mismatch, it has successfully implemented A! This strengthens the taxonomy rather than defeating it.

Meta-Admissibility & The Infinite Regress

The Objection: If Ω can be revised, why not D ? And if D , why not the rules governing D ?



Transition
rules (Γ)

Admissibility
spaces (D)

Hypotheses (Ω)

The Reality: This is not a vicious regress; each level answers a fundamentally different question.

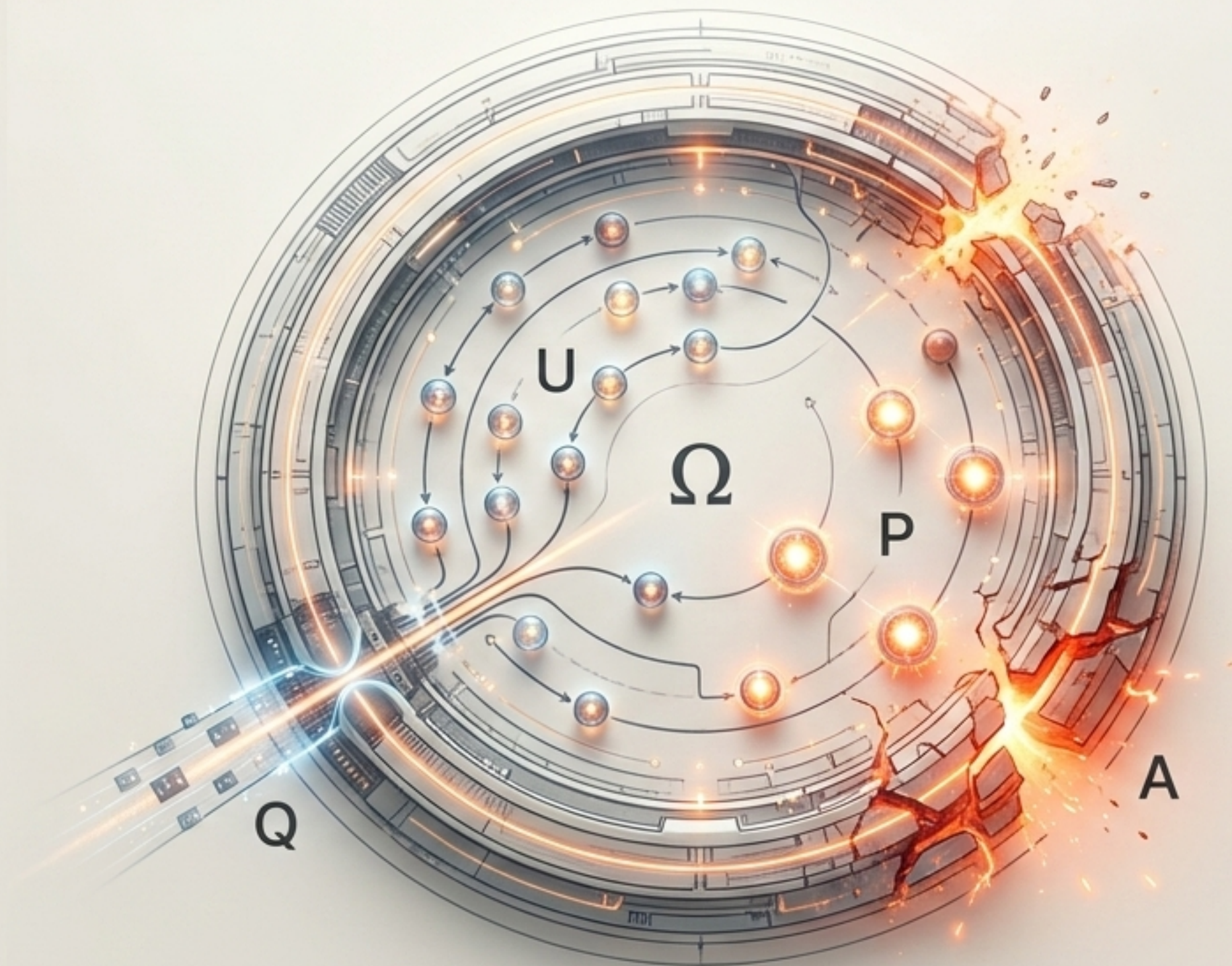
Layer 1 (Ω): The space of live hypotheses.

Layer 2 (D): What counts as a fixable admissibility space.

Layer 3 (Γ): What counts as a fixable rule for changing D .

Conclusion: We don't need a theory of meta-admissibility (Γ) to prove that revising a hypothesis space (D) is distinct from revising a hypothesis (Ω).

Synthesis: Redefining Computational Boundaries



The Expanded Cycle

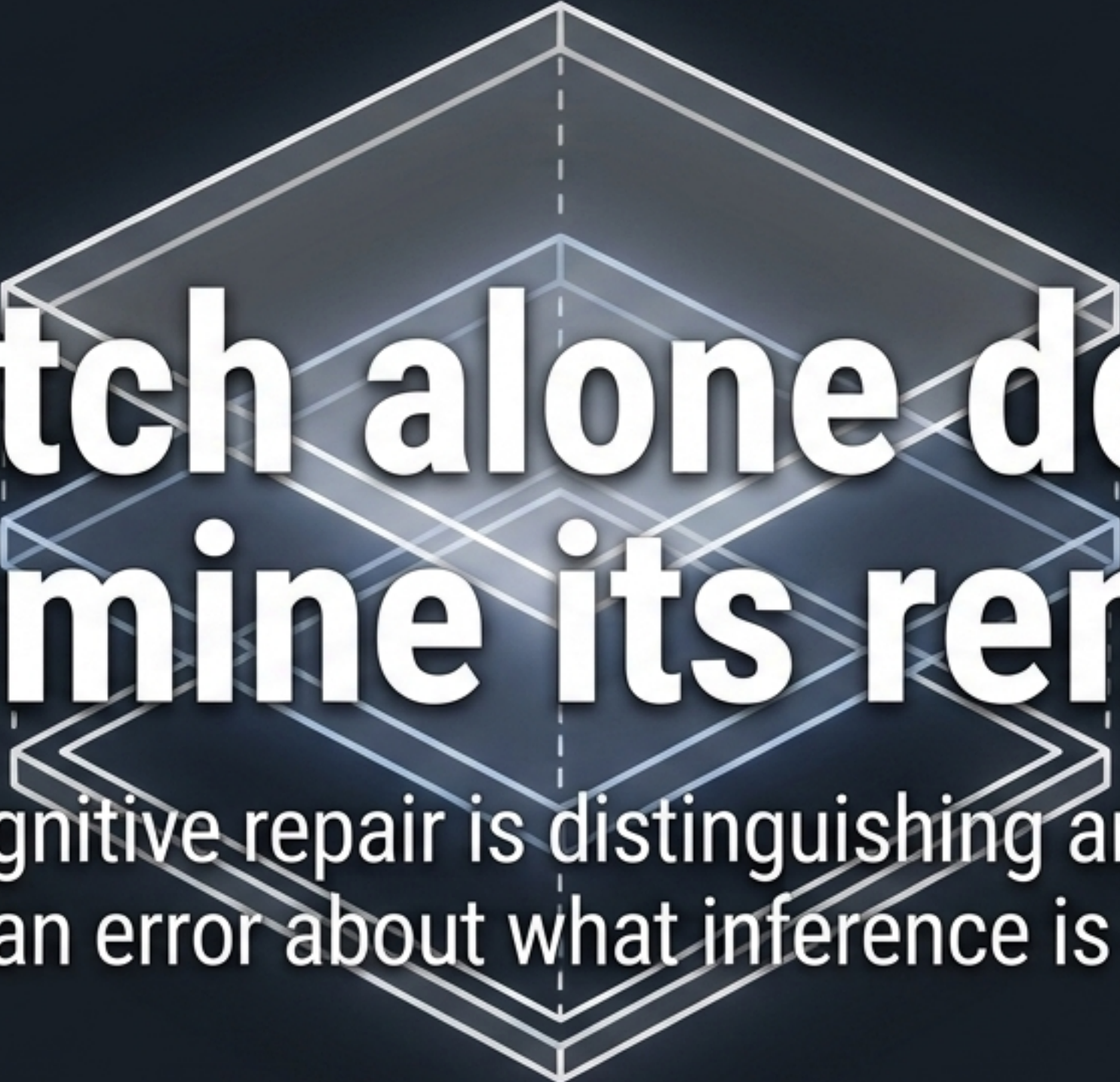
A complete inferential system incorporates U, Q, P, and A acting upon bounded state spaces.

The Role of Boundaries

Boundaries do not merely screen statistical states; they determine what counts as a possible state, transition, or continuation.

The Final Rule

Before asking how an error propagates across a boundary, we must ask if the error concerns a state within the space, or the constitution of the space.



Mismatch alone does not determine its remedy.

The first step of cognitive repair is distinguishing an error for inference to correct from an error about what inference is permitted to be.

Inference within a space is not equivalent to revision of the space.
(Flyxion, 2026)